STAT2004 – Tutorial Week 8

We will go through Exercises 1–2 together in the tutorials.

Exercise 1 (Pivot transformation for exponential data)

Let $X \sim \exp(\lambda)$, with pdf and cdf given respectively by

$$f_\lambda(x) = \lambda e^{-\lambda x} \quad \text{and} \quad F_\lambda(x) = 1 - e^{-\lambda x} \ , \quad x \geq 0 \ .$$

Here, $\lambda > 0$ is an unknown parameter.

(a) Find the distribution of $Y = \lambda X$ and show that it is a pivot quantity.

(b) Find cutoff values a and b such that the following probability statement holds: $99\% = P(a \leq Y \leq b)$.

(c) Using your results from (a) and (b), or otherwise, construct a 99% confidence interval for λ based on the observation X.

(d) Evaluate the confidence interval from part (c) based on the realisation $x = 12.41882$. Interpret your confidence interval in the form of a sentence.

(e) The true value of λ that generated the realisation $x = 12.41882$ is hidden somewhere on Alan's laptop. Did the interval you calculated in part (d) contain this true value?

Exercise 2 (Prior and posterior beliefs)

In Week 7 Lecture 3, Max stated that his prior belief about the true proportion p of Skittles *before* we conducted our in-class experiment was "around 50%" . What if our prior belief was the opposite of Max's?

(a) Consider a prior belief about p that is characterized by

$$f(p) \propto p^{0.5-1}(1-p)^{0.5-1} \ ,$$

which corresponds to a Beta(0.5, 0.5) distribution. Graph this prior belief $f(p)$ as a function of p across the range $0 \leq p \leq 1$.

(b) Using the graph from part (a), qualitatively describe this prior belief. In particular, explain why this is (in some sense) the "opposite" of Max's belief.

(c) Show that after observing $x = 23$ Skittles out of $n = 47$ samples, our posterior belief should be updated to

$$f(p|x = 23) \propto p^{23+0.5-1}(1-p)^{24+0.5-1} \ .$$

What distribution does this correspond to?

(d) Graph the posterior belief $f(p|x = 23)$ as a function of p across the range $0 \leq p \leq 1$.

(e) Give an intuitive explanation as to why our posterior belief here has changed so much from our prior belief.